

17-803 Empirical Methods

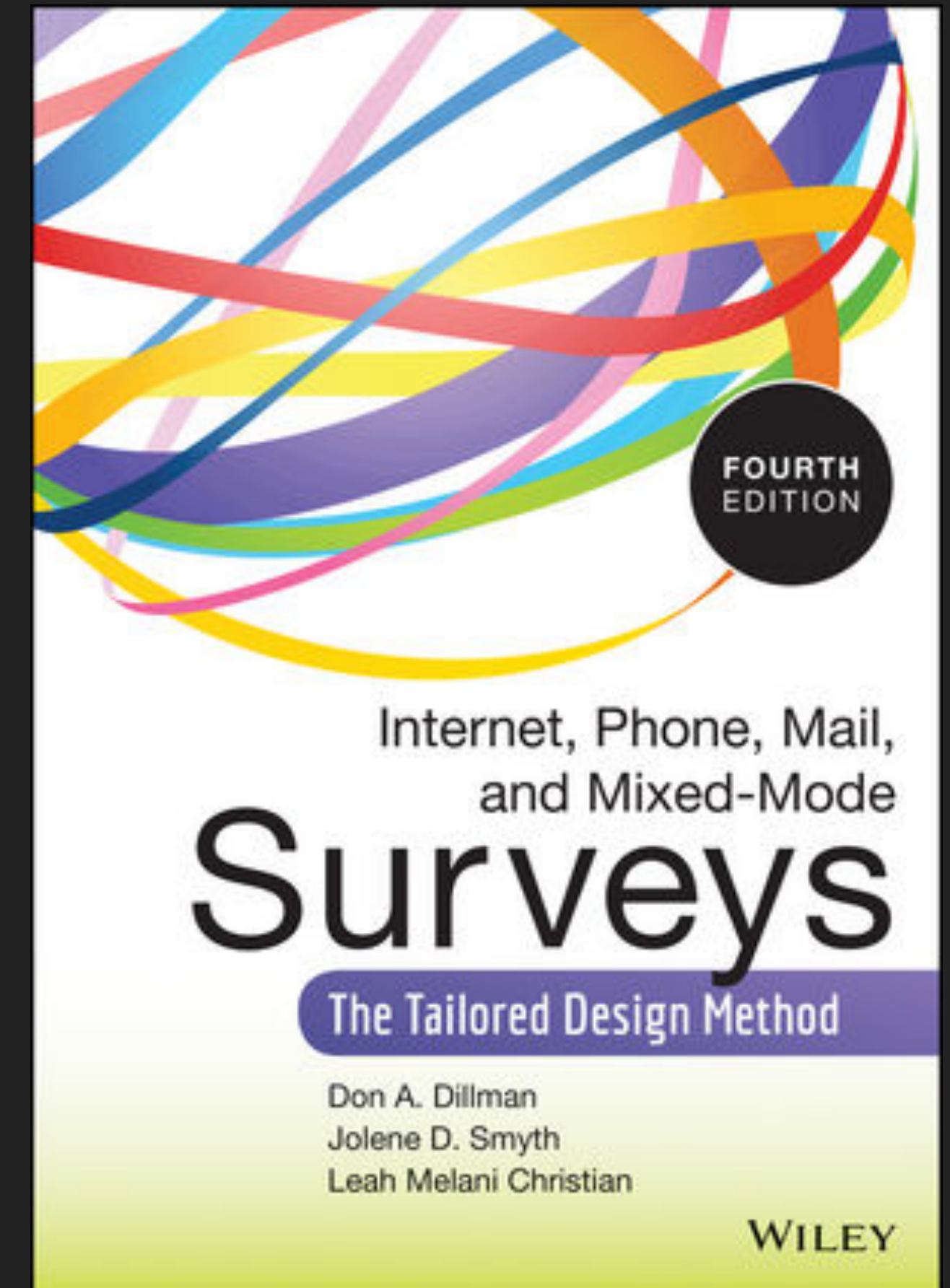
Bogdan Vasilescu, S3D

Survey Design (Part 1)

Friday, February 20, 2026

Outline for Today

- ▶ Chapters from Dillman, D., Smyth, J. D., & Christian, L. M. (2014). Internet, Phone, Mail and Mixed-Mode Surveys: The Tailored Design Method (4th ed.). Hoboken, NJ: Wiley.
 - ▶ Ch1: Sample Surveys in our Electronic World
 - ▶ Ch2: Reducing People's Reluctance to Respond to Surveys
 - ▶ Ch4: The Fundamentals of Writing Questions
 - ▶ Ch5: How to Write Open and Closed Ended Questions
- ▶ Hof, M. (2012). Questionnaire Evaluation with Factor Analysis and Cronbach's Alpha. Student project. Seminar in Methodology and Statistics. Uni. Groningen
- ▶ Yong, A. G., & Pearce, S. (2013). A beginner's guide to factor analysis: Focusing on exploratory factor analysis. *Tutorials in quantitative methods for psychology*, 9(2), 79-94.
- ▶ Cairns, P. (2019). Doing better statistics in human-computer interaction. Cambridge University Press.
 - ▶ Ch15: What Makes a Good Likert Item?
 - ▶ Ch16: The Meaning of Factors
 - ▶ Ch17: Unreliable Reliability: The Problem of Cronbach's Alpha
 - ▶ Ch18: Tests for Questionnaires



What Is a Survey Good for?

- ▶ Prevalence (sort of)
- ▶ Relations between variables
- ▶ Differences among subpopulations

Probability sample survey strength:

Collect data from only a sample of the population but generalize results to the whole

Main goal when designing probability sample surveys:

Reduce survey error (the difference between an estimate that is produced using survey data and the true value of the variables in the population that one hopes to describe)

Four Types of Error To Minimize To Improve the Survey Estimates

- ▶ Sampling frame – how to construct?
 - ▶ Coverage error
- ▶ Draw sample – probability sampling?
 - ▶ Sampling error
- ▶ Administer survey – who responded?
 - ▶ Nonresponse error
- ▶ Questions as measures – valid and reliable?
 - ▶ Measurement error

Coverage Error Example

- ▶ Landline random digit dial telephone survey
 - ▶ People who have landlines are quite different from those who do not on a number of important characteristics (e.g., higher socioeconomic status).
- ▶ Context matters
 - ▶ Internet survey among US population vs CMU students



"Butler c. 1922" CC-BY-SA-4.0 Wikipedia

Sampling Error

- ▶ Surprising how few people you would need to survey to obtain estimates with acceptable levels of precision!
- ▶ CMU: 14k students
 - ▶ 95% confidence +/- 2% margin of error:
 - ▶ Random sample of 2050 students
- ▶ US population: 328 million
 - ▶ 95% confidence +/- 2% margin of error:
 - ▶ Random sample of 2384 people
 - ▶ 95% confidence +/- 10% margin of error:
 - ▶ Random sample of 96 people

Sample Size Calculator

This Sample Size Calculator is presented as a public service of Creative Research Systems [survey software](#). You can use it to determine how many people you need to interview in order to get results that reflect the target population as precisely as needed. You can also find the level of precision you have in an existing sample.

Before using the sample size calculator, there are two terms that you need to know. These are: **confidence interval** and **confidence level**. If you are not familiar with these terms, [click here](#). To learn more about the factors that affect the size of confidence intervals, [click here](#).

Enter your choices in a calculator below to find the sample size you need or the confidence interval you have. Leave the Population box blank, if the population is very large or unknown.

Determine Sample Size

Confidence Level: ☒ 95% ☐ 99%

Confidence Interval:

Population:

Sample size needed:

Find Confidence Interval

Confidence Level: ☒ 95% ☐ 99%

Sample Size:

Population:

Percentage:

Confidence Interval:

<https://www.surveysystem.com/sscalc.htm>

Aside: Confidence Level, Confidence Interval

- ▶ **Confidence interval** (aka "margin of error"):
 - ▶ Plus-or-minus figure usually reported
 - ▶ Lower is better
 - ▶ Example:
 - ▶ if confidence interval is 4 and 47% percent of sample answer X
 - ▶ then you can be "sure" that among the entire relevant population between 43% (47-4) and 51% (47+4) would have answered X.
- ▶ **Confidence level** (how sure can you be):
 - ▶ Represents how often the true percentage of the population who would answer X lies within the confidence interval.
 - ▶ Higher is better
 - ▶ Common: 95% or 99%

Nonresponse Error

- ▶ Do the characteristics of respondents differ from those who chose not to respond in a way that is relevant to the study results?
- ▶ Higher response rates reduce the likelihood of nonresponse error.
 - ▶ But: nonresponse error may occur in surveys with higher as well as lower response rates (Groves & Peytcheva, 2008)

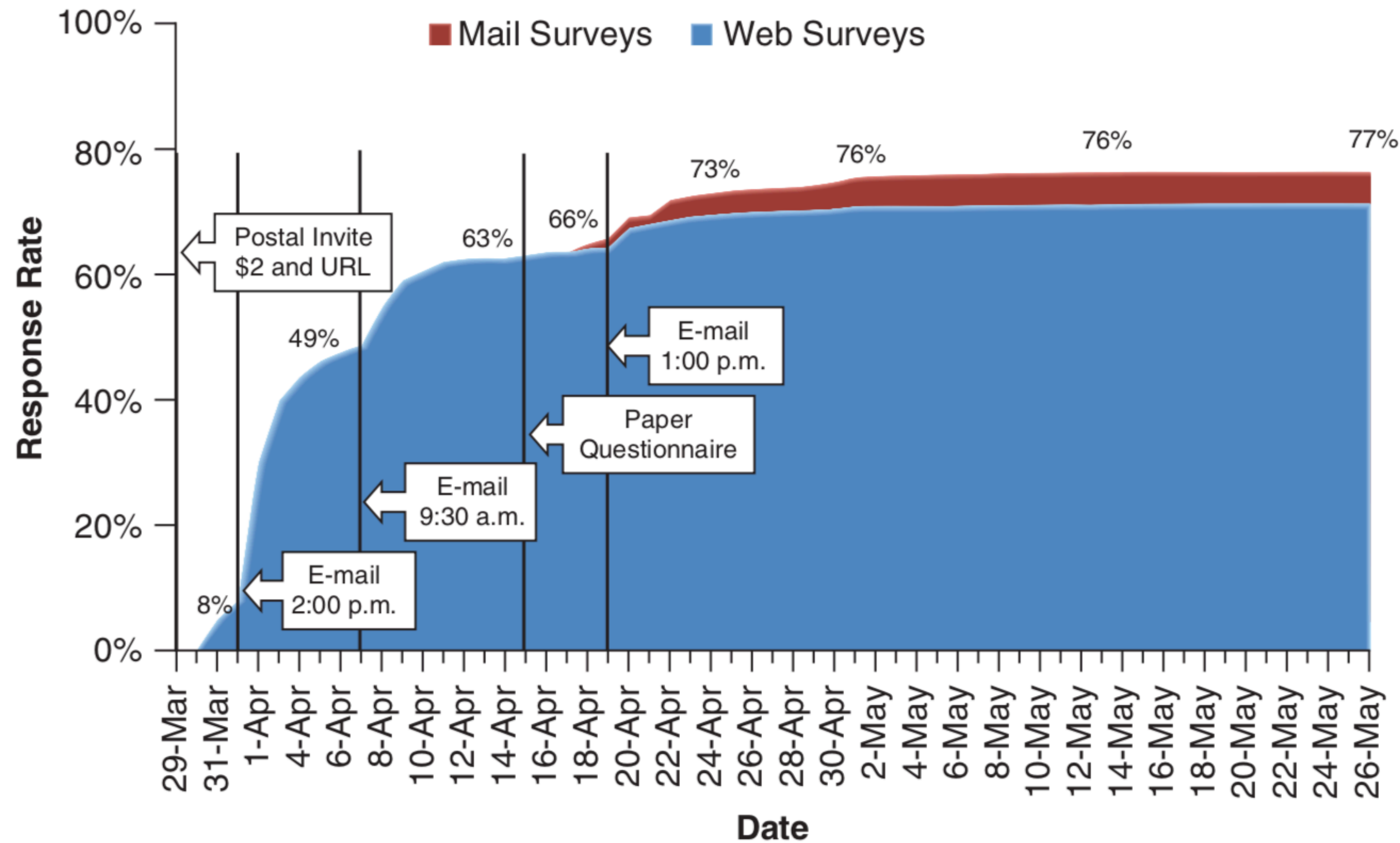
Measurement Error

- ▶ Asking good questions:
 - ▶ Does question adequately measure the idea or concept of interest?
 - ▶ “Construct validity”
- ▶ Example:
 - ▶ household income for the previous year as a measure of household wealth.
 - ▶ annual income decreases after retirement, but wealth may not.

Getting people to respond:

Example survey of 600 PhD students at Washington State U.
about their dissertation work and graduate training

FIGURE 2.2 Cumulative response rate by day and mode for the 2013 WSU Doctoral Student Experience Survey, showing contribution of each contact to final response rate.



Source: Adapted from *Determining Whether Research Is Interdisciplinary: An Analysis of New Indicators* (Technical Report 13-049), by M. M. Millar, 2013, Pullman: Washington State University, Social and Economic Sciences Research Center.

Initial commu- nication

Jane Doe
123 Cougar Road
Pullman, WA 99163

Dear Jane,

I am writing to ask for your help with an important survey we are conducting of WSU doctoral students. I understand that you have successfully completed your preliminary examinations and are now at the stage of needing to complete a dissertation.

My colleague, Morgan Millar, and I have been working with the National Science Foundation to better understand how the needs of doctoral training in the U.S. are changing. We are hoping you could spend a few minutes sharing some of your experiences in your doctoral program. In particular, we are interested in factors that may affect bringing the dissertation writing process to a successful conclusion.

To this end, we would greatly appreciate if you would answer a few questions for us. To do so, simply go to this website: www.opinion.wsu.edu/phdexperience

In order to begin the survey, you will need to enter this access code: «RESPID».

We think it should only take about ten minutes to complete the questionnaire.

The survey is confidential. Your individual answers will not be linked with your name or department in any reports of the data. Your participation is voluntary and if you come to any question you prefer not to answer, you are welcomed to skip it and go on to the next. Should you have any questions or comments, please contact me (dillman@wsu.edu) or Thom Allen, the study director (509) 335-1722 or ted@wsu.edu.

We very much appreciate your help with this study, and a small token of appreciation is enclosed with this letter as way of saying thank you.

Many Thanks,



Don A. Dillman
Regents Professor and Deputy Director

Reminder

Dear Jane,

Earlier this week we sent you a letter asking for your help with an important survey. We are conducting this study of WSU doctoral students to learn more about the processes they go through to complete their dissertations and finish their degrees.

I am following up with this e-mail to provide you with an electronic link to the survey website. I hope this link makes it easier for you to respond. It should only take a few minutes to complete the questionnaire.

Simply click on this link and you will automatically be logged into the survey:

<http://www.opinion.wsu.edu/phdexperience>

And enter your personal Access Code in the space provided: <<RESPID>>

The results of this study will help us better understand the needs and experiences of students as they work on their dissertation research. Your participation is very important, and we appreciate you considering our request.

Sincerely,



Don A. Dillman
Regents' Professor and Deputy Director
Social and Economic Sciences Research Center

The Social Reward Value of Modest Incentives With the Request (Avdeyeva & Matland, 2013)

- ▶ Control group – no incentive
- ▶ A second group – 50 rubles (~\$1.65) with survey request
- ▶ A third group – promised 300 rubles if questionnaire was returned
- ▶ Response rates:
 - ▶ no incentive – 10%
 - ▶ 50 rubles pre pay – 37%
 - ▶ 300 rubles post pay – 24%
 - ▶ Combined pre and post pay – 48%

Using social exchange concepts to motivate potential respondents:

People are more likely to comply with a request from someone else if they believe and trust that the rewards for complying with that request will eventually exceed the costs of complying

Increasing the Benefits of Survey Participation

- ▶ Make it seem valuable, important
 - ▶ How will results be useful?
 - ▶ Ask for help/advice
 - ▶ Make it interesting
 - ▶ Gamification
 - ▶ Scarcity – only some people get asked
 - ▶ Others have responded
 - ▶ Pay (a little bit) forward

Decreasing the Costs of Participation

- ▶ Make it easy
 - ▶ Keep it short!
 - ▶ Keep it simple and clear and convenient
 - ▶ Giving people a choice of modes may reduce response rates
- ▶ Be likable
 - ▶ Show respect
 - ▶ Don't use subordinate language

Establishing Trust

- ▶ Worry about malware, fake surveys
- ▶ Ways to assess your trustworthiness (contact info)
- ▶ Sponsorship
- ▶ Assure confidentiality
- ▶ Minimize requests to obtain personal information

FIGURE 2.3 Summary of ways to increase benefits, reduce costs, and establish trust.



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Survey Design (Part 2)

Friday, February 20, 2026

Issues to consider when drafting questions

Social Desirability Bias

- ▶ Wanting to make a good impression
- ▶ “How often do you drive a car after drinking alcoholic beverages? Frequently, Occasionally, Seldom, Never, or Don’t Know.” (Dillman & Tarnai, 1991)
 - ▶ interviewer-administered telephone survey – 63% said “never”
 - ▶ self-administered paper survey – 52% responded “never”
- ▶ “How would you describe your current health? Excellent, Good, Fair, or Poor.”
 - ▶ fewer respondents choose “excellent” in self-administered surveys compared to interviews (Biemer, 1997; Hochstim, 1967).
 - ▶ conventional American greeting “How are you?” → always “Fine”

Social Desirability Bias

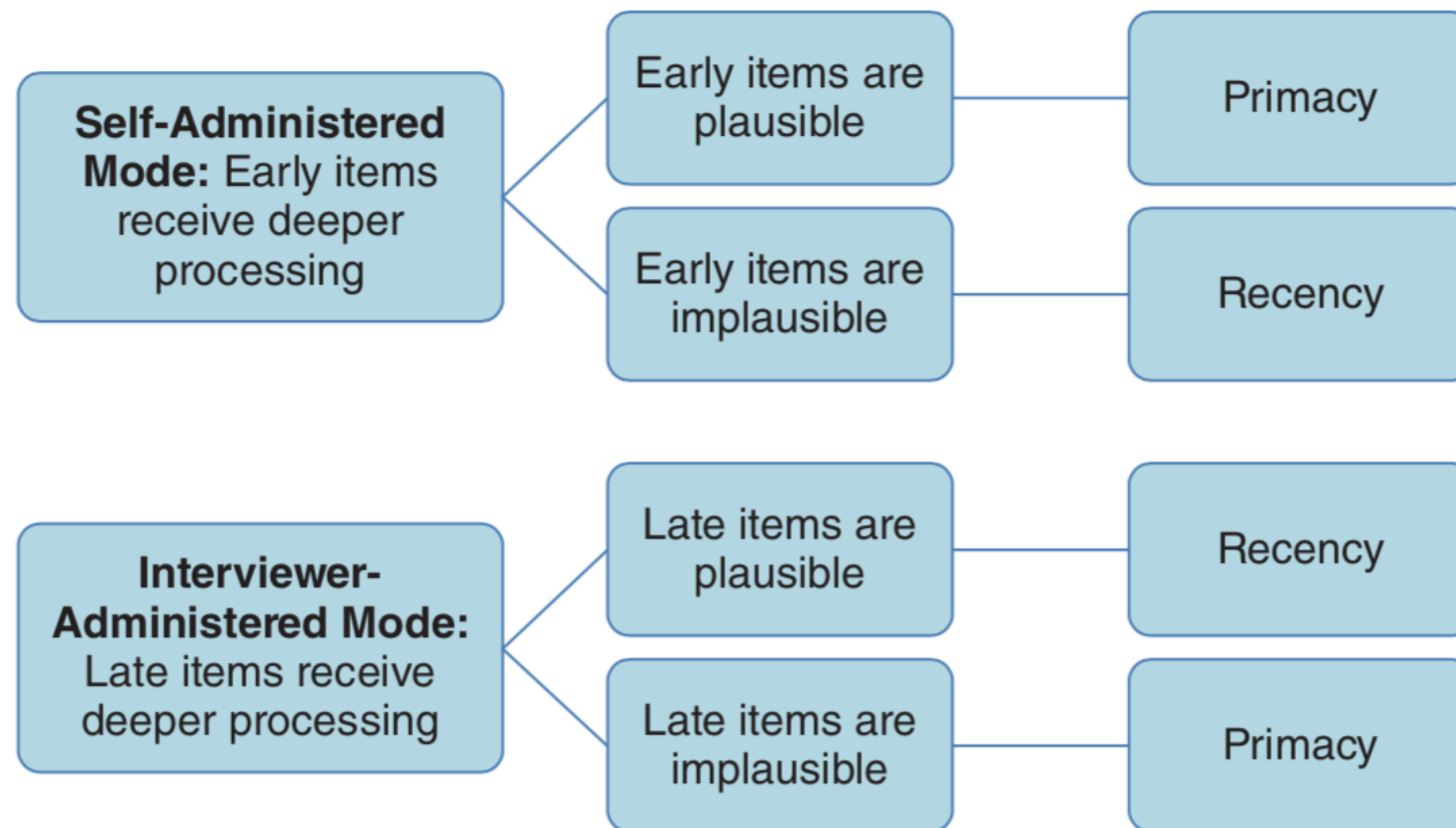
- ▶ Kreuter et al (2008):
 - ▶ 20% of web survey respondents denied having ever received a D or F in college when in fact they had received one of these grades.
- ▶ Catania et al (1996):
 - ▶ both men and women were more likely to report engaging in extramarital sex when interviewed by a same sex interviewer than when interviewed by an opposite sex interviewer.

Acquiescence

- ▶ The tendency to agree with someone rather than disagree
- ▶ Schuman and Presser (1981):
 - ▶ 60% agreed "Individuals are more to blame than social conditions for crime and lawlessness in this country."
 - ▶ 57% of a control group agreed with the exact reverse "Social conditions are more to blame than individuals for crime and lawlessness in this country."

Primacy and Recency

FIGURE 4.4 Cognitive elaboration model of response order effects.



Source: Adapted from "A Cognitive Model of Response Order Effects in Survey Measurement," by N. Schwarz, H. J. Hippler, and E. Noelle-Neumann, 1992, in N. Schwarz and S. Sudman (Eds.), *Context Effects in Social and Psychological Research* (pp. 187–201), New York, NY: Springer-Verlag.

Tendency to more frequently choose from among the first / last categories offered regardless of their content

Anchoring

- ▶ Occurs when an early response option forms a standard of comparison for later response options.
- ▶ Noelle-Neumann (1970): "Which food is more typically German?"
 - ▶ "potatoes" and "rice" – 30% said "potatoes"
 - ▶ "rice" and "potatoes" – 48% said "potatoes"



The anatomy of a survey question

Types of Survey Questions

Open-ended questions

Question stem **What is the most important problem facing Nebraska today?**

Answer space



*Question stem
with verbal and
numeric
instructions*

How many years have you lived in Nebraska?

Please report only whole numbers. For example, if you have lived in Nebraska 20 months, please round to 2 years.

*Answer space
with verbal and
symbolic
instruction*



Years lived in Nebraska

Closed-ended ordinal question

Question stem **Overall, how satisfied are you with living in Nebraska?**

Answer choices

- ☐ Completely satisfied
- ☐ Very satisfied
- ☐ Somewhat satisfied
- ☐ Not too satisfied
- ☐ Not at all satisfied

Types of Survey Questions

Body Height Reported by U.S. Men

As part of a comprehensive health survey, the U.S. CDC asked roughly 200,000 adult men in 2021 this question: "About how tall are you without shoes?"

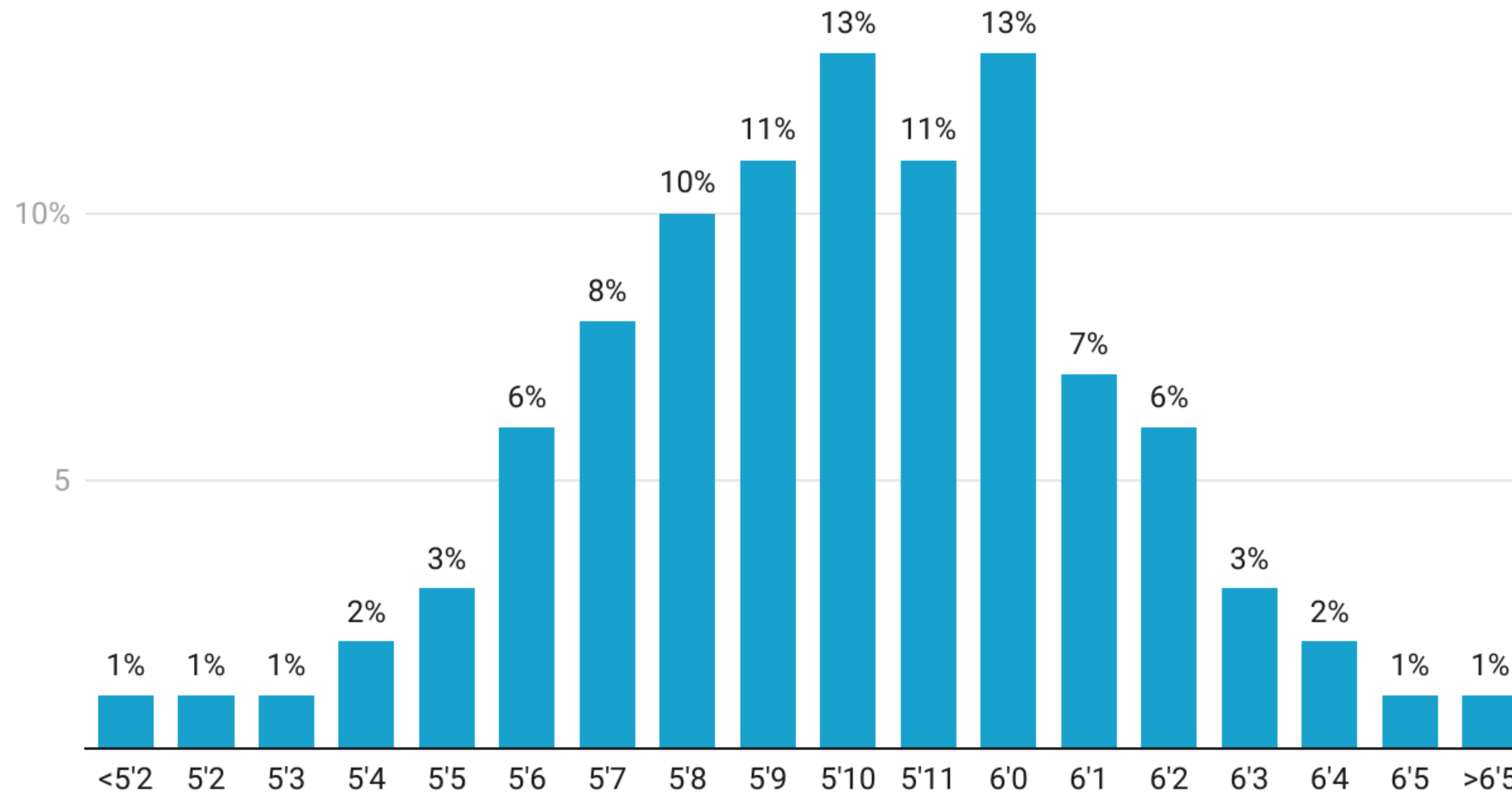


Chart: u/academiaadvice • Source: CDC

Types of Survey Questions

Closed-ended nominal question

Question stem **What is your current marital status?**

- Answer choices*
- ☐ Married
 - ☐ Living with a partner
 - ☐ Divorced
 - ☐ Separated
 - ☐ Widowed
 - ☐ Never married

Partially closed-ended question

Question stem **What are your favorite women's sports at the University of Nebraska?**

- Answer choices*
- ☐ Basketball
 - ☐ Cross Country
 - ☐ Gymnastics
 - ☐ Soccer
 - ☐ Softball
 - ☐ Swimming and Diving
 - ☐ Tennis
 - ☐ Volleyball
 - ☐ Other: Please specify

Use Specific and Concrete Words To Specify the Concepts Clearly

- ▶ Questions may be factual, but interesting ones usually involve concepts
- ▶ What are the concepts?

FIGURE 4.11 Use specific and concrete words to specify the concepts clearly.

Question with vague concepts

How many times did you eat together as a family last week?

☐ Number of meals

A revised question with more specific and concrete concepts

How many meals did you eat together as a family at home last week?

☐ Number of meals

A more specific revision

How many meals did you sit down to eat at home as a family last week?

☐ Number of meals

- ▶ Use previously-validated scales where possible

Guidelines for Forming Questions

- ▶ Choose the appropriate question format
- ▶ Make sure the question applies to the respondent
- ▶ Ask one question at a time
- ▶ Make sure the question is technically accurate
- ▶ Use simple and familiar words
- ▶ Use specific and concrete words to specify the concepts clearly
- ▶ Use as few words as possible to pose the question
- ▶ Use complete sentences that take a question form, and use simple sentence structures
- ▶ Make sure “yes” means yes and “no” means no
- ▶ Organize questions to make it easier for respondents to comprehend the response task

How to Write Open- and Closed-Ended Questions

Open Ended Questions – Wording Specificity

FIGURE 5.1 How specificity of question wording affects reports about when students began their studies.

Question Wording (Telephone)	% Reporting Month and Year	% Reporting Season/Semester
<u>When</u> did you begin your studies at Washington State University?	13.4	57.3
<u>What date</u> did you begin your studies at Washington State University?	49.5	32.3
<u>What month and year</u> did you begin your studies at Washington State University?	83.7	11.0

Source: "Helping Respondents Get It Right the First Time: The Influence of Words, Symbols, and Graphics in Web Surveys," by L. M. Christian, D. A. Dillman, and J. D. Smyth, 2007b, *Public Opinion Quarterly*, 71(1), pp. 113–125.

Open Ended Questions — Provide Extra Motivation To Respond

- ▶ “In your own words, how would you describe your adviser(s)?”

Vs

- ▶ “This question is very important to understanding the Washington State University student experience. Please take your time answering it.

In your own words, how would you describe your adviser(s)?”

Open Ended Questions – Use Nondirective Probes To Obtain More Information

- ▶ Smyth et al. (2007b) – “What businesses would you most like to see in the Pullman and Moscow area that are currently not available?”
 - ▶ A random half of students received a follow-up probe asking, “Are there any others?”
- ▶ No probe: average of 1.8 businesses
- ▶ Probe: average of 2.4 businesses

Open Ended Questions – the Type of Probe Used Will Strongly Impact the Amount and Type of Information Received

- ▶ Smyth et al. (2006) – “In your own words, how would you describe your adviser or advisers?”
- ▶ Probes:
 - ▶ “Is there anything else?” – 18% responded; most said “no”
 - ▶ “Can you tell me more about that?” – 82% responded with additional information, including new ideas or themes as well as elaboration on previously reported themes

Closed Ended Questions – Acquiescence

Poor Designs

Do you favor congressional term limits of four years?

- ☐ Favor
- ☐ Oppose

How satisfied are you with the overall service you have received from your financial consultant?

- ☐ Very satisfied
- ☐ Somewhat satisfied
- ☐ Somewhat dissatisfied
- ☐ Very dissatisfied

Closed Ended Questions – Acquiescence

FIGURE 5.4 State both positive and negative sides in the question stem.

<u>Poor Designs</u>	<u>Improved Designs</u>
Do you favor congressional term limits of four years? ☐ Favor ☐ Oppose	Do you favor or oppose congressional term limits of four years? ☐ Favor ☐ Oppose
How satisfied are you with the overall service you have received from your financial consultant? ☐ Very satisfied ☐ Somewhat satisfied ☐ Somewhat dissatisfied ☐ Very dissatisfied	How satisfied or dissatisfied are you with the overall service you have received from your financial consultant? ☐ Very satisfied ☐ Somewhat satisfied ☐ Somewhat dissatisfied ☐ Very dissatisfied

Closed Ended Questions – Primacy

FIGURE 5.7 Subtraction effects in multiple-answer questions.

Which of the following resources have you used at WSU? Please check all that apply.			
<u>Original Order</u>	<u>%</u> <u>Endorsing</u>	<u>Reverse Order</u>	<u>%</u> <u>Endorsing</u>
Libraries	95	Counseling Services	
Computer Labs		Library Instruction	52
Student Health Center		Campus-Sponsored Tutoring	
Academic Advising		Career Services	
Student Recreation Center		Internet/E-Mail Access	
Internet/E-Mail Access		Student Recreation Center	
Career Services		Academic Advising	
Campus-Sponsored Tutoring		Student Health Center	
Library Instruction	20	Computer Labs	
Counseling Services		Libraries	93

Closed Ended Questions – Forced Choice

Check-all-that-apply formatted question

Which of the following items do you have? Please check all that apply.

- ☐ Desktop computer
- ☐ Laptop computer
- ☐ Cell phone
- ☐ E-reader
- ☐ Tablet computer
- ☐ iPod or MP3 player

Closed Ended Questions – Forced Choice

FIGURE 5.12 Replacing check-all-that-apply questions with a forced-choice format.

Check-all-that-apply formatted question

Which of the following items do you have? Please check all that apply.

- ☐ Desktop computer
- ☐ Laptop computer
- ☐ Cell phone
- ☐ E-reader
- ☐ Tablet computer
- ☐ iPod or MP3 player

A revision converting to the forced-choice format

Do you have each of the following items or not?

- | Yes | No | |
|-----------------------|-----------------------|--------------------|
| ☐ | ☐ | Desktop computer |
| ☐ | ☐ | Laptop computer |
| ☐ | ☐ | Cell phone |
| ☐ | ☐ | E-reader |
| ☐ | ☐ | Tablet computer |
| ☐ | ☐ | iPod or MP3 player |

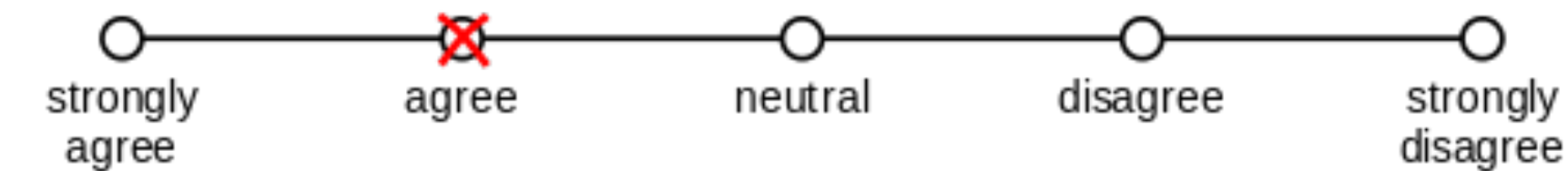
Understanding Likert scales better

Likert Scale

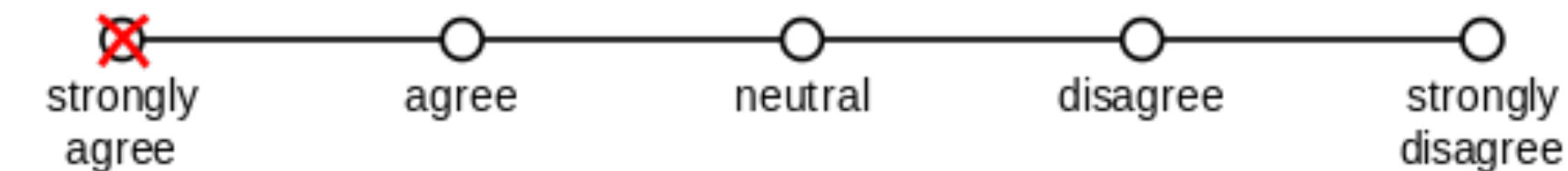
(Pronounced Lick-Ert)

Website User Survey

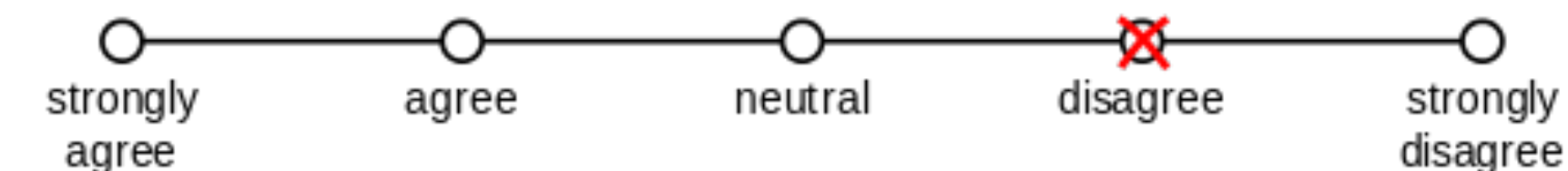
1. The website has a user friendly interface.



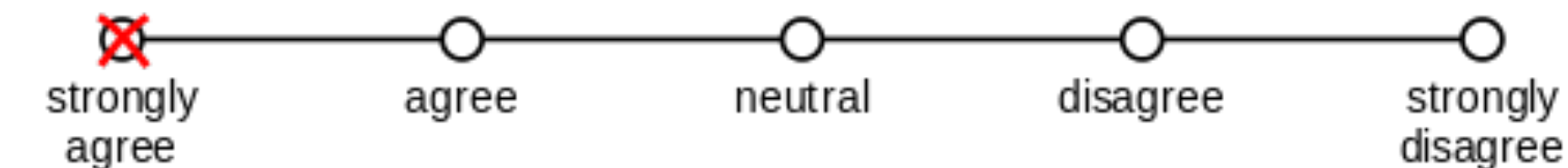
2. The website is easy to navigate.



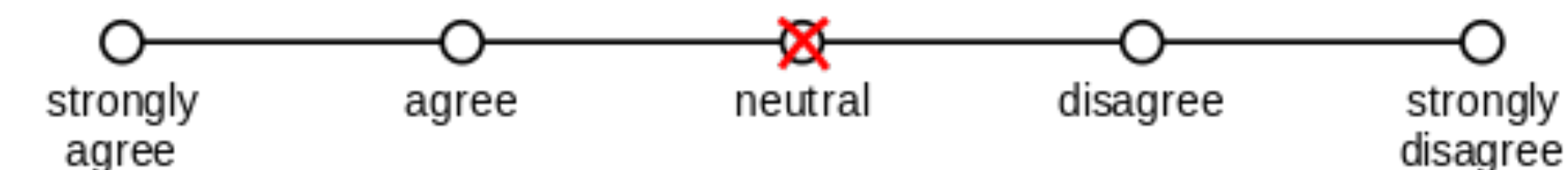
3. The website's pages generally have good images.



4. The website allows users to upload pictures easily.



5. The website has a pleasing color scheme.



"Example Likert Scale using five Likert Items
pertaining to Wikipedia" CC-BY-SA-3.0 Wikipedia

Likert Scale (Pronounced Lick-Ert)

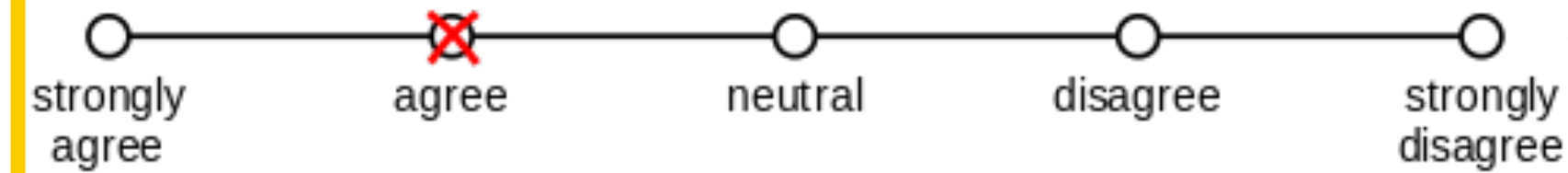
“strongly disagree” = 1

...

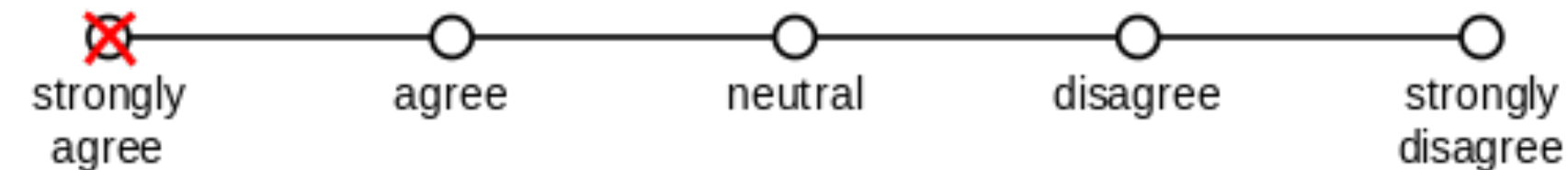
“strongly agree” = 5

Website User Survey

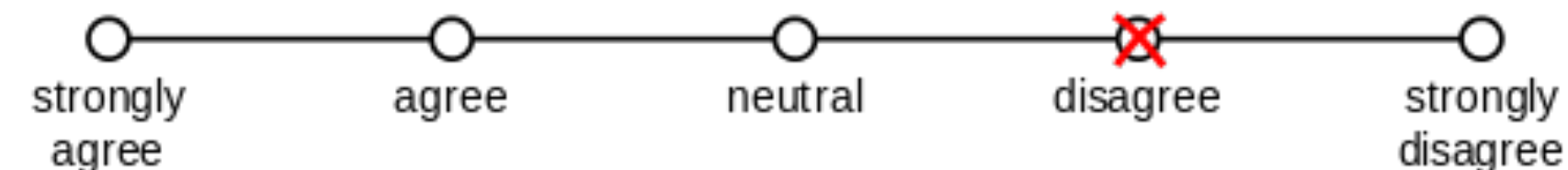
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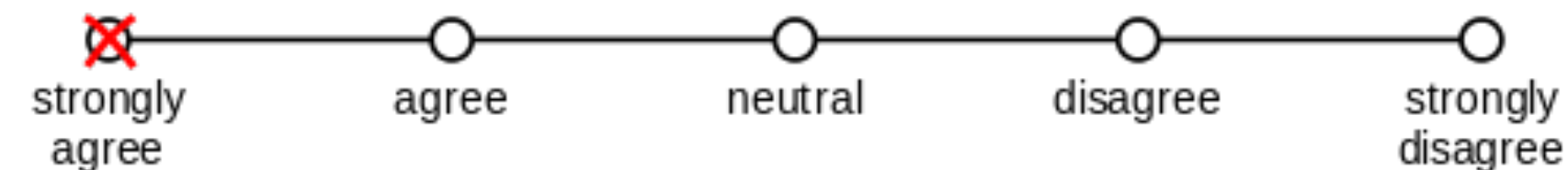
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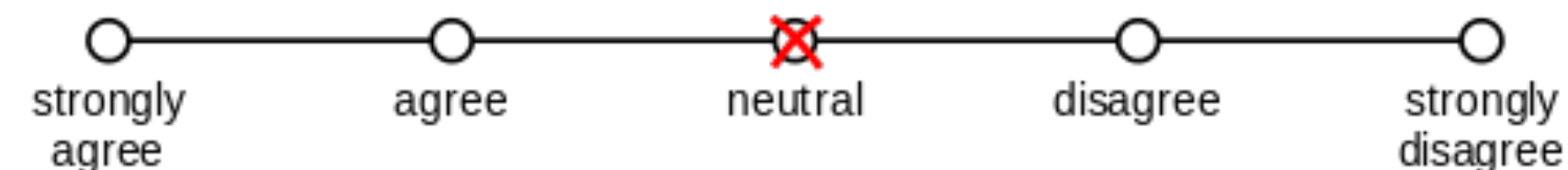
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Likert item

Likert Scale

“Example Likert Scale using five Likert Items
pertaining to Wikipedia” CC-BY-SA-3.0 Wikipedia

Concerns: Reliability – Is Whatever Is Being Measured Reliably Found?

- ▶ Statistical or internal reliability (aka “**consistency**”):
 - ▶ The extent to which the items of a scale are consistent with each other.
 - ▶ Typically evaluated using Cronbach’s α
 - ▶ A good scale should have a good split-half correlation.
 - ▶ Cronbach’s α is effectively the average of all the possible split-half correlations.
- ▶ Reliability over time (aka “**stability**”)
 - ▶ Over time, people should answer the same items the same way.
 - ▶ Especially for stable attributes like attitude or personality

Concerns: Validity – Does Whatever Is Being Measured Correspond to the Concept That We Say It Does?

- ▶ Do scales with different response formats agree with each other?
 - ▶ e.g., similar mean values with different items
- ▶ Do items correlate with reference criteria of what the concept means?

Should Items Have a Midpoint?

1. I like massively multiplayer online role-playing games (MMORPG)

Strongly
Disagree

Disagree

Neither

Agree

Strongly
Agree

Should Items Have a Midpoint?

Really ambivalent about their attitude or simply do not have an opinion?

1. I like massively multiplayer online role-playing games (MMORPG)
Strongly Disagree Disagree Neither Agree Strongly Agree
2. I like massively multiplayer online role-playing games (MMORPG)
Strongly Disagree Disagree Agree Strongly Agree
3. I like massively multiplayer online role-playing games (MMORPG)
Strongly Disagree Disagree Agree Strongly Agree No Opinion

Figure 15.2 Three formats of Likert item: 1. with midpoint; 2. no midpoint; 3. no midpoint but option to have no opinion

One Problem of Having a Midpoint Is That of Acquiescence Bias

- ▶ A neutral option means respondents can comfortably avoid disagreeing even if they do actually disagree.
- ▶ Garland (1991) experiment:
 - ▶ 4-point (no midpoint) vs 5-point items
 - ▶ scale scores were higher for the 5-point items
 - ▶ But Nadler et al (2015) found no statistically significant differences

But Bias Could Also Occur in the Opposite Direction

- ▶ 4-point item: Forcing people to respond in a way that they did not truly feel would introduce an element of randomness into the answers.

Likert Item Design Summary

- ▶ Likert items are pretty robust to variations in response format.
 - ▶ Midpoint vs not
 - ▶ 5 vs 7 options
 - ▶ Strongly disagree, Disagree, Neither, Agree, Strongly agree
 - ▶ Strongly disagree, Moderately disagree, Slightly disagree, Neither, Slightly agree, Moderately agree, Strongly agree
 - ▶ All-labeled vs just end-points labeled
 - ▶ Strongly Disagree, Disagree, Neither, Agree, Strongly Agree
 - ▶ Strongly Disagree, 1, 2, 3, 4, 5, Strongly Agree
- ▶ Use larger scales over single items.

Example: Questionnaire to Seventh' Graders Pleasure in Writing

- ▶ I love writing.
- ▶ Writing is my favorite school subject.
- ▶ When I write, I feel well.
- ▶ I hate writing.
- ▶ I write as soon as I get the chance.
- ▶ I make sure that I have to write as less as possible.
- ▶ I write more than my class mates.
- ▶ When I write, I prefer to do something different.
- ▶ Writing gives me pleasure.
- ▶ I just write, when I can get a good grade for it.
- ▶ Writing is boring.
- ▶ I like different kinds of writing.
- ▶ When I have the opportunity to determine on my own what I do in the Dutch class, I usual do a writing task.
- ▶ I write even if the teacher does not assign a writing task.
- ▶ I would like to spend more time on writing.
- ▶ Writing is a waste of time.
- ▶ I always look forward to writing lessons.
- ▶ I write because I have to at school.
- ▶ I like it to write down my thoughts.
- ▶ I would like to write more at school.

Example: Questionnaire to Seventh' Graders Pleasure in Writing

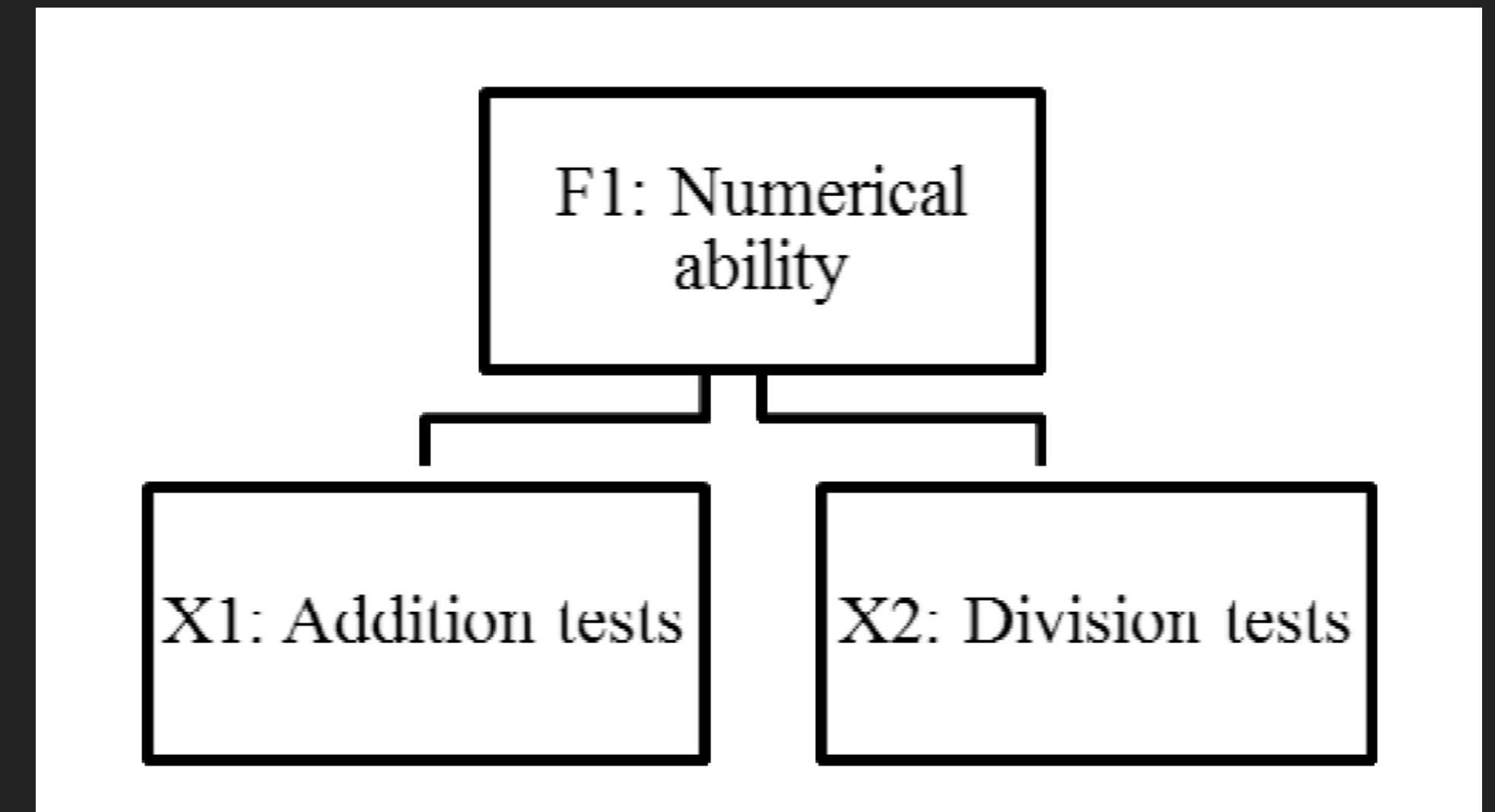
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- ▶ I write because I have to at school.
- ▶ I like it to write down my thoughts.
- ▶ I would like to write more at school.

Example: Questionnaire to Seventh' Graders Pleasure in Writing

- ▶ I love writing.
- ▶ Writing is my favorite school subject.
- ▶ When I write, I feel well.
- ▶ I hate writing.
- ▶ I write as soon as I get the chance.
- ▶ I make sure that I have to write as less as possible.
- ▶ I write more than my class mates.
- ▶ When I write, I prefer to do something different.
- ▶ Writing gives me pleasure.
- ▶ I just write, when I can get a good grade for it.
- ▶ Writing is boring.
- ▶ I like different kinds of writing.
- ▶ When I have the opportunity to determine on my own what I do in the Dutch class, I usual do a writing task.
- ▶ I write even if the teacher does not assign a writing task.
- ▶ I would like to spend more time on writing.
- ▶ Writing is a waste of time.
- ▶ I always look forward to writing lessons.
- ▶ I write because I have to at school.
- ▶ I like it to write down my thoughts.
- ▶ I would like to write more at school.

Factor Analysis To Assess Construct Validity

- ▶ Factor analysis:
 - ▶ A statistical technique for explaining the variance in a set of observed variables through a smaller number of unobserved *latent* variables called factors.
 - ▶ If many variables move together, they probably share an underlying cause – e.g., scores on addition and division tests may reflect a common "numerical ability" factor.
 - ▶ The model says each observed variable is a linear combination of a few common factors plus a unique error term. The goal is to find those factors and quantify how strongly each variable loads onto them.



In This Context

- ▶ Exploratory factor analysis detects the constructs - i.e. factors - that underlie a dataset based on the correlations between questionnaire items.
- ▶ The factors that explain the highest proportion of variance the questionnaire items share are expected to represent the underlying constructs.

The Correlation Matrix and its Eigenvectors

- ▶ You start with the correlation matrix R of your variables. Its eigenvectors (and eigenvalues) come from the standard decomposition $R = Q\Lambda Q^T$, where Q contains the eigenvectors as columns and Λ is a diagonal matrix of eigenvalues.
- ▶ Each eigenvector defines a direction in the space of your variables – a weighted combination of variables that captures a particular dimension of variation.
- ▶ The corresponding eigenvalue tells you how much total variance that direction accounts for. A large eigenvalue means the associated eigenvector points along a direction of strong common variance; a small eigenvalue (near zero) means that direction contributes little.

The Correlation Matrix and its Eigenvectors

- ▶ The eigenvectors with the largest eigenvalues become your factors.
- ▶ The first eigenvector points in the direction of maximum variance across all variables – it's the single linear combination that explains the most.
- ▶ The second eigenvector is orthogonal to the first and explains the next most variance, and so on.

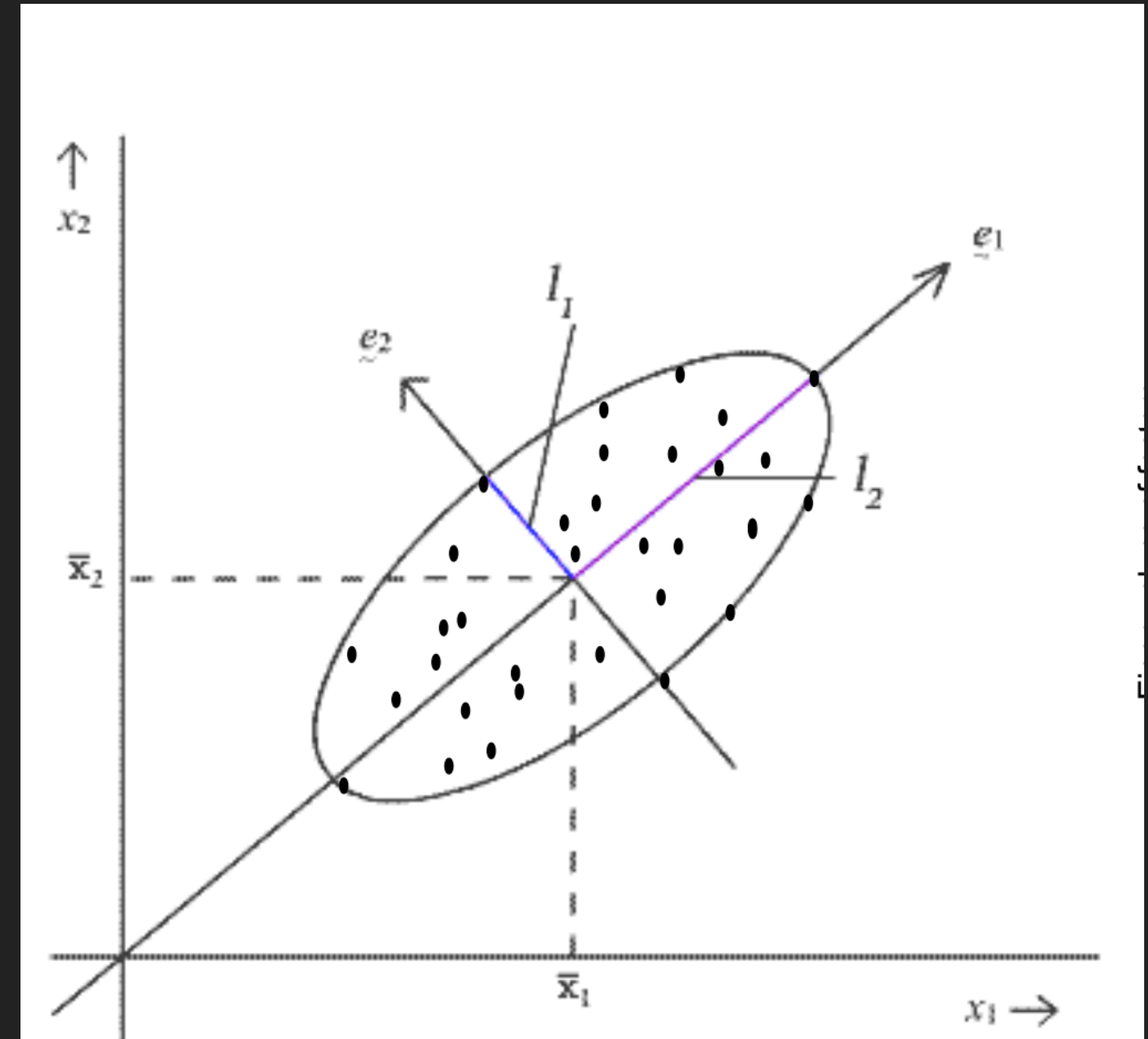


Figure 2. Scatterplot of two variables (x_1 and x_2). The lines e_1 and e_2 represent the eigenvectors of the correlation matrix of variables x_1 and x_2 . The eigenvalue of an eigenvector is the length of an eigenvector measured from one end of the oval to the other end.

Dimensionality Reduction: Keep Only the Eigenvectors With Eigenvalues Above Some Threshold (Eigenvalue > 1 , the "Kaiser Criterion")

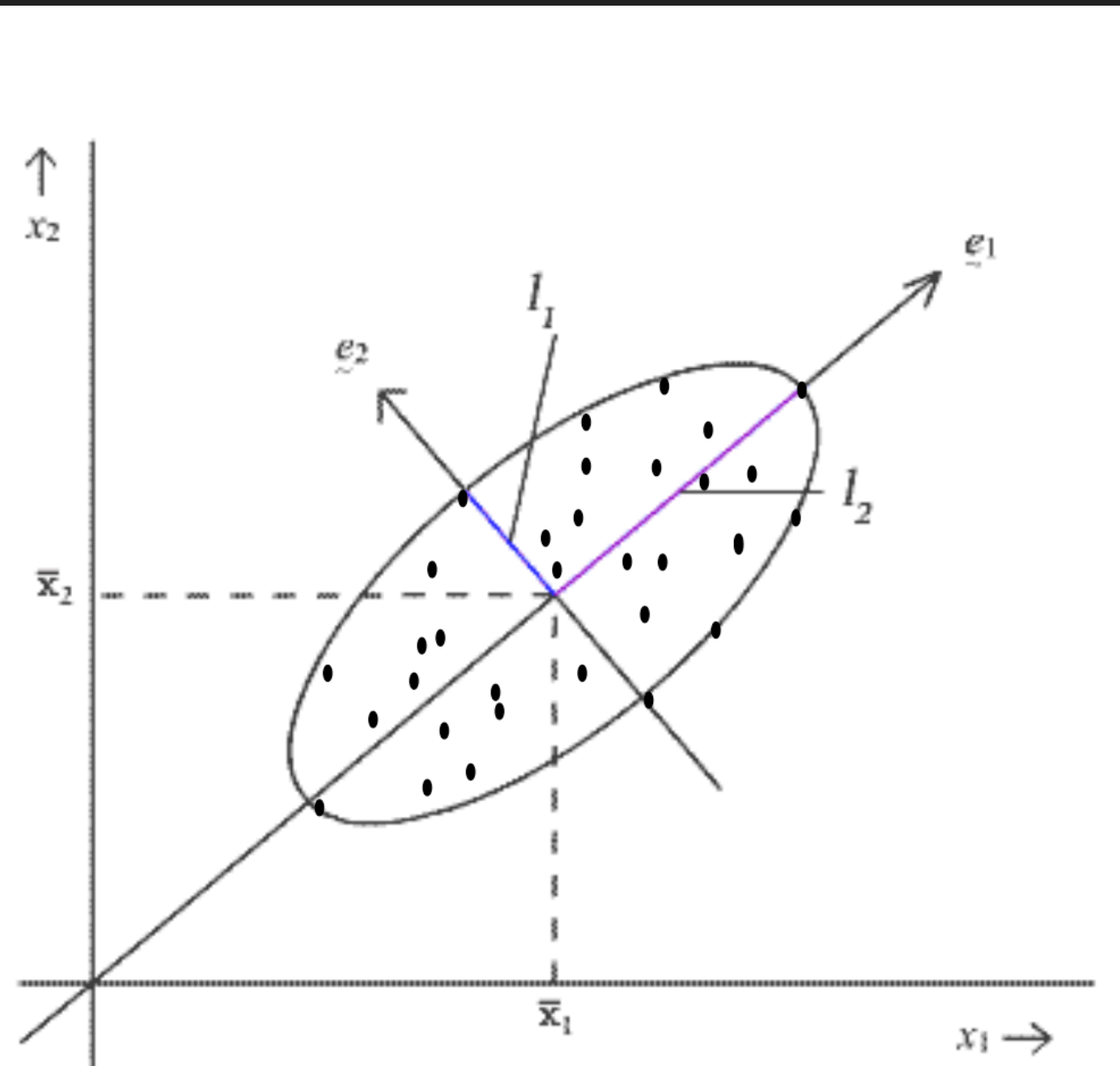


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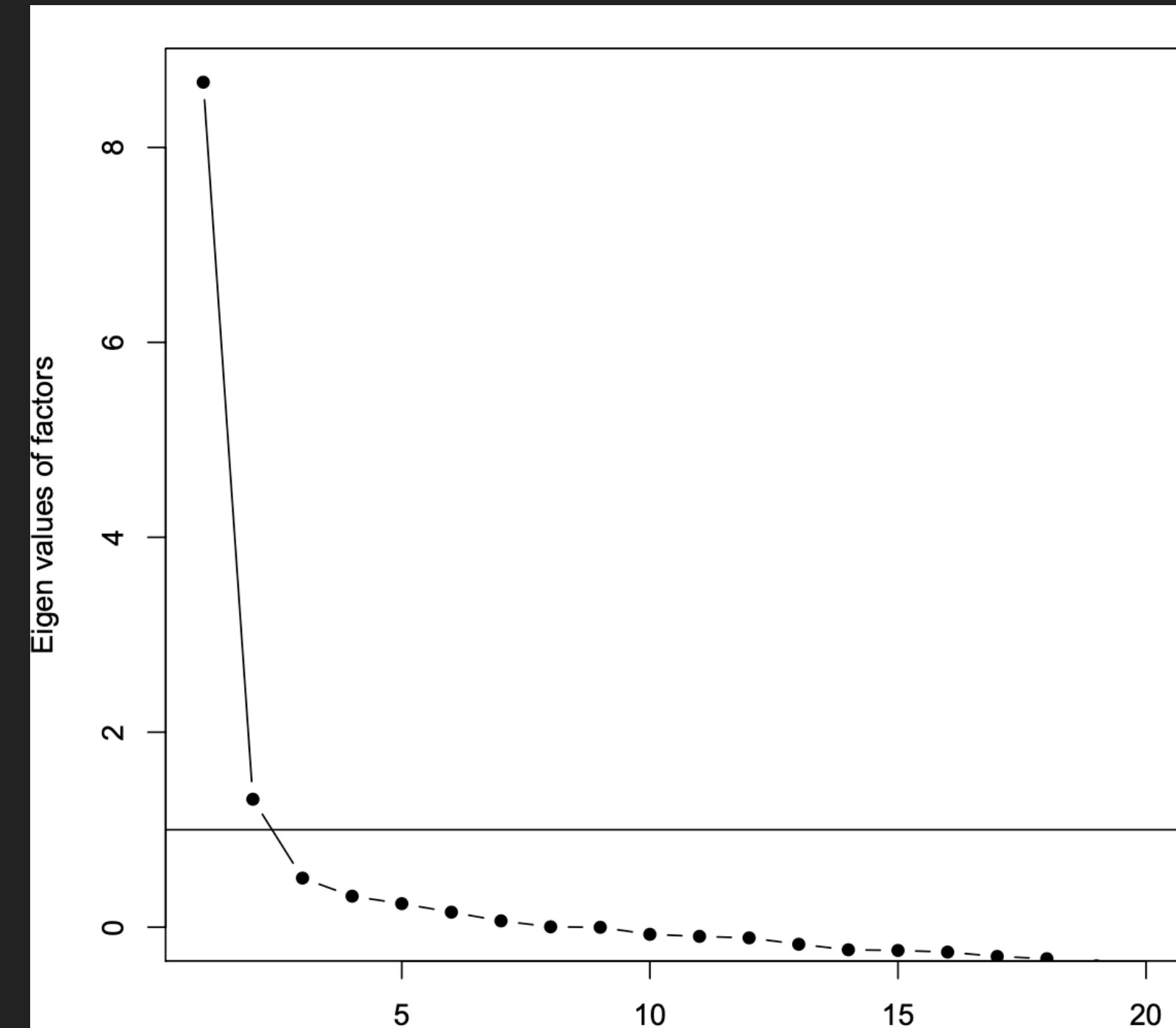


Figure 3. Screeplot of factors underlying the dataset. Every point represents one factor.

Rotation (Like Varimax) Helps Make the Factor Loadings More Interpretable

- ▶ After extraction, your factors are mathematically optimal but often hard to interpret – most variables load moderately on most factors.
- ▶ Rotation reorients the factor axes in the factor space to achieve **simple structure**: each variable loads highly on one factor and near-zero on the others.
- ▶ Rotation doesn't change the total variance explained or how well the model fits. It only redistributes the variance among factors to make them more meaningful.

Example Rotation

- ▶ Suppose you give 300 students 6 tests and extract 2 factors. Before rotation, your loadings look like this:

Test	Factor 1	Factor 2
Vocabulary	0.75	0.50
Reading comprehension	0.70	0.45
Verbal reasoning	0.72	0.48
Arithmetic	0.68	-0.52
Algebra	0.65	-0.55
Geometry	0.60	-0.50

Note: every test loads substantially on Factor 1, and Factor 2 splits the verbal/math tests by sign.

Example Rotation

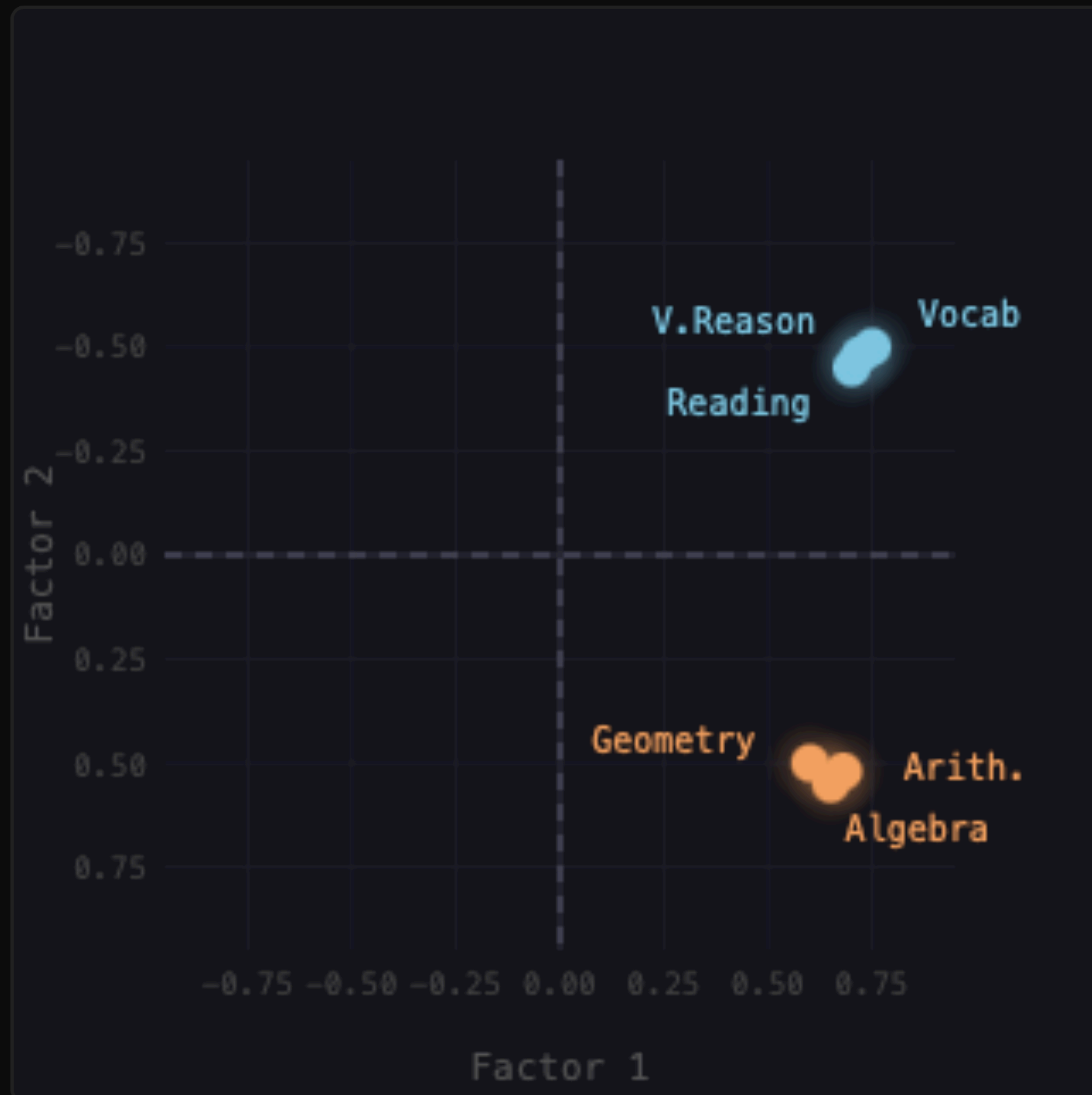
► After varimax rotation, the same two factors become:

Test	Factor 1	Factor 2
Vocabulary	0.88	0.08
Reading comprehension	0.82	0.10
Verbal reasoning	0.85	0.09
Arithmetic	0.07	0.85
Algebra	0.06	0.88
Geometry	0.08	0.79

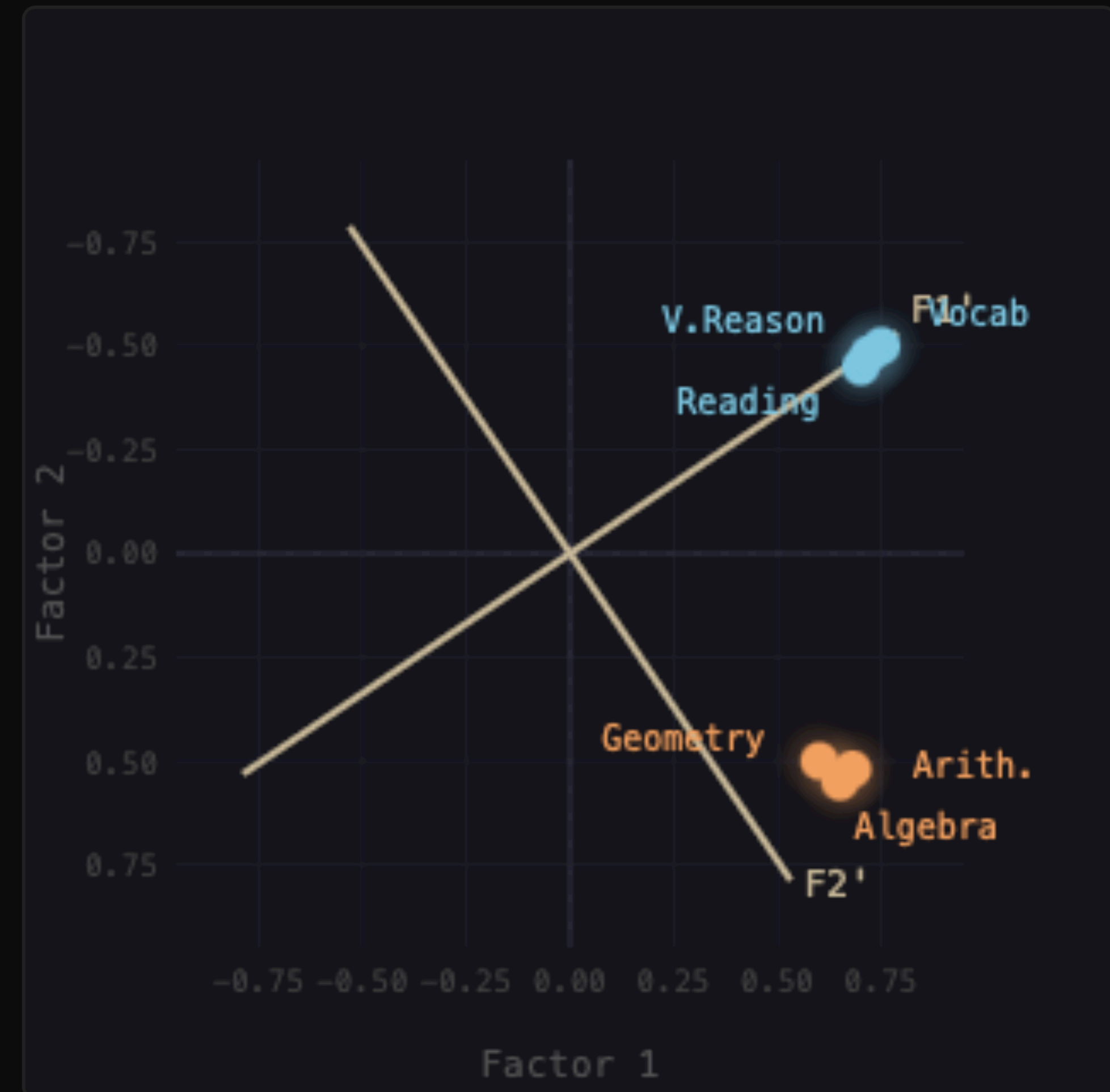
Now Factor 1 is clearly verbal ability and Factor 2 is clearly mathematical ability.

Example Rotation

BEFORE ROTATION — UNROTATED



AFTER ROTATION — VARIMAX



● Verbal tests ● Math tests ---- Unrotated axes — Rotated axes

Orthogonal Vs. Oblique Rotation

- ▶ **Varimax** is orthogonal – it keeps factors uncorrelated, i.e., the axes stay perpendicular. This is clean and simple but sometimes unrealistic (verbal and math ability probably are somewhat correlated in the real world).
- ▶ **Oblique** rotation (e.g., promax, oblimin) allows the axes to tilt toward each other, permitting correlated factors. This often gives even cleaner simple structure when the underlying constructs are naturally related, but you get an extra matrix to interpret (the factor correlation matrix).
- ▶ The choice between them is partly empirical (does oblique rotation reveal strong factor correlations?) and partly theoretical (does your domain knowledge suggest the factors should be independent?).

The Positively Formulated Items in This Questionnaire Make Up the First Factor and the Negatively Formulated Items the Second Factor.

- ▶ p01. I love writing.
- ▶ p02. Writing is my favorite school subject.
- ▶ p03. When I write, I feel well.
- ▶ p04. I hate writing.
- ▶ p05. I write as soon as I get the chance.
- ▶ p06. I make sure that I have to write as less as possible.
- ▶ p07. I write more than my class mates.
- ▶ p08. When I write, I prefer to do something different.
- ▶ p09. Writing gives me pleasure.
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- ▶ p20. I would like to write more at school.

```
Call:
factanal(x = na.omit(passion), factors = 2, rotation = "oblimin")

Loadings:
      Factor1 Factor2
p01  0.547  -0.289
p02  0.747
p03  0.727
p04      0.802
p05  0.625
p06      0.641
p07  0.463
p08 -0.133  0.558
p09  0.702 -0.115
p10      0.597
p11      0.680
p12  0.412 -0.243
p13  0.719  0.122
p14  0.739
p15  0.758
p16      0.771
p17  0.917
p18      0.739
p19  0.623
p20  0.771

SS loadings      Factor1 Factor2
Proportion Var   6.141    3.534
Cumulative Var   0.307    0.177
Cumulative Var   0.307    0.484

Factor Correlations:
      Factor1 Factor2
Factor1  1.000  -0.642
Factor2 -0.642   1.000

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 197.76 on 151 degrees of freedom.
The p-value is 0.00636
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```

Common pattern that reverse-phrased items load on a different factor (Schmitt & Stults, 1985)

Cronbach's Alpha Analysis

Reliability analysis					
Call: alpha(x = passion)					
	alpha	average_r	mean	sd	
	0.93	0.42	3.3	0.36	
Reliability if an item is dropped:			Item statistics		
	alpha	average_r	n	r	r.cor mean sd
p01	0.93	0.41	p01 114	0.77	0.76 3.5 1.19
p02	0.93	0.42	p02 114	0.68	0.67 3.9 1.07
p03	0.93	0.41	p03 114	0.77	0.76 3.6 1.06
p04-	0.93	0.42	p04- 114	0.70	0.69 3.3 1.29
p05	0.93	0.43	p05 113	0.57	0.55 3.8 0.90
p06-	0.93	0.42	p06- 114	0.61	0.59 3.0 1.19
p07	0.93	0.43	p07 114	0.55	0.52 3.4 0.82
p08-	0.93	0.42	p08- 114	0.63	0.60 2.6 1.18
p09	0.93	0.41	p09 114	0.78	0.77 3.6 0.98
p10-	0.93	0.43	p10- 114	0.59	0.57 2.5 1.23
p11-	0.93	0.42	p11- 114	0.60	0.58 3.1 1.26
p12	0.93	0.42	p12 114	0.62	0.59 3.0 1.00
p13	0.93	0.42	p13 114	0.60	0.57 3.4 1.08
p14	0.93	0.42	p14 114	0.71	0.70 3.7 1.01
p15	0.93	0.42	p15 114	0.71	0.70 3.8 0.95
p16-	0.93	0.42	p16- 113	0.63	0.62 3.4 1.12
p17	0.93	0.41	p17 113	0.80	0.80 3.8 0.85
p18-	0.93	0.42	p18- 114	0.62	0.61 2.6 1.17
p19	0.93	0.42	p19 114	0.67	0.66 3.2 1.06
p20	0.93	0.41	p20 114	0.77	0.76 3.8 0.94

Table 4. Output of a Cronbach's alpha analysis in R

Example: Multi-Item Scale for Brainstorming

1. The group aimed to generate as many ideas as possible.
2. All ideas were welcome, no matter how unconventional they were.
3. The group tried to combine similar ideas into one.
4. The group aimed to build on the ideas generated.
5. Ideas were generated first individually, then discussed as a group.
6. An organizer or group leader facilitated brainstorming for my session/group.
7. Group members criticized ideas proposed during the group/session. (R)

Responses on 5-point Likert scale, asked to what extent the statement reflected the way their group decided to work, from “not at all” to “completely.”

Analyzed inter-item reliability (Cronbach's α), had to drop #7

From: Filippova, A., Trainer, E., & Herbsleb, J. D. (2017). *From diversity by numbers to diversity as process: supporting inclusiveness in software development teams with brainstorming*. Paper presented at the International Conference on Software Engineering, Buenos Aires, Argentina.

Survey Examples

- ▶ Ecosystem survey

http://cmu.ca1.qualtrics.com/jfe/form/SV_d4M66VwPlZYd5kh

- ▶ Results: <http://breakingapis.org/survey/>

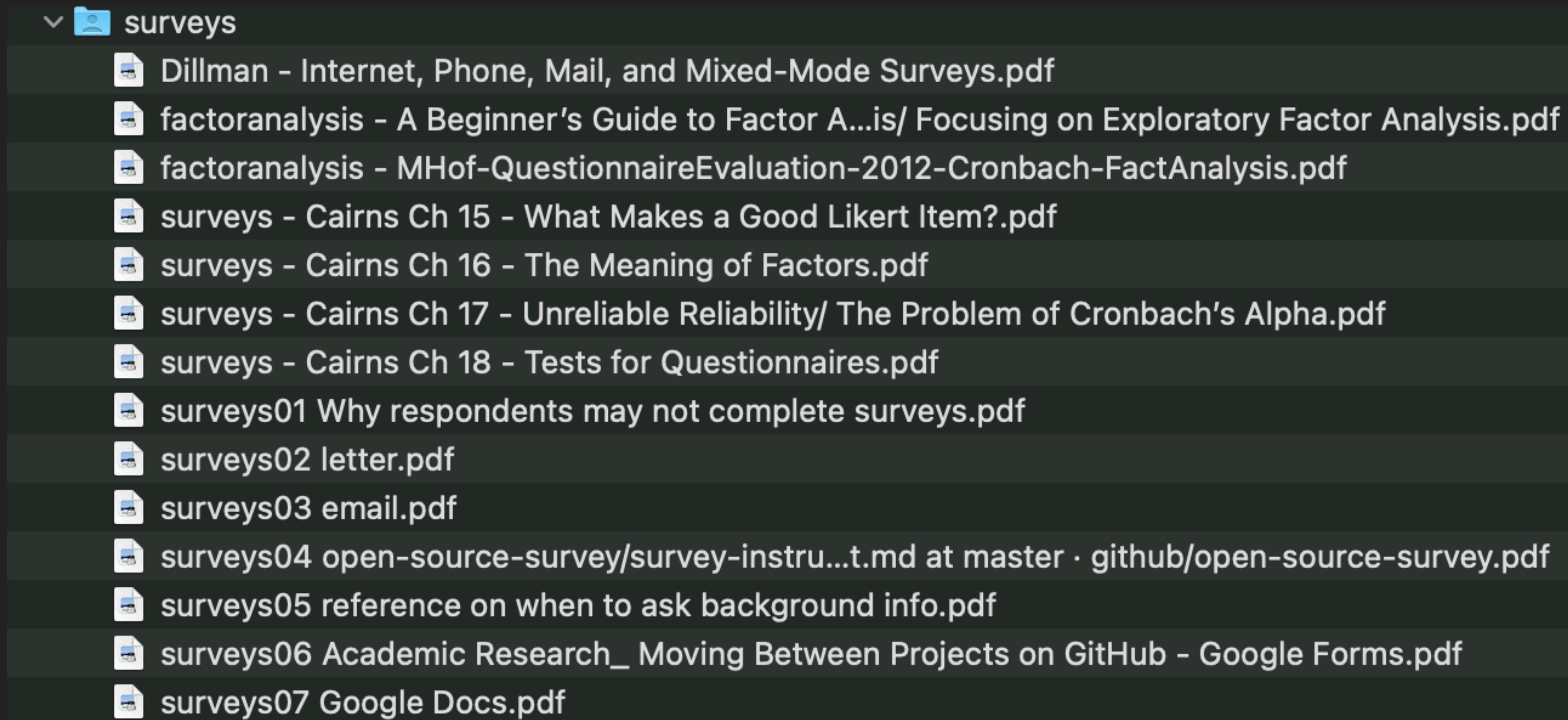
- ▶ GitHub open source survey

<https://github.com/github/open-source-survey>

- ▶ Results: <http://opensourcesurvey.org/2017/>

Readings

- ▶ https://drive.google.com/drive/folders/1ISOQlbw-cRmT47_itpIJkTZua_IASNgB?usp=sharing



Credits

▶ Graphics:

- ▶ Dave DiCello photography (cover)

▶ Content:

- ▶ Chapters from Dillman, D., Smyth, J. D., & Christian, L. M. (2014). *Internet, Phone, Mail and Mixed-Mode Surveys: The Tailored Design Method* (4th ed.). Hoboken, NJ: Wiley.
 - ▶ Ch1: Sample Surveys in our Electronic World
 - ▶ Ch2: Reducing People's Reluctance to Respond to Surveys
 - ▶ Ch4: The Fundamentals of Writing Questions
 - ▶ Ch5: How to Write Open and Closed Ended Questions
- ▶ Hof, M. (2012). *Questionnaire Evaluation with Factor Analysis and Cronbach's Alpha*. Student project. Seminar in Methodology and Statistics. University of Groningen
- ▶ Yong, A. G., & Pearce, S. (2013). A beginner's guide to factor analysis: Focusing on exploratory factor analysis. *Tutorials in quantitative methods for psychology*, 9(2), 79-94.
- ▶ Cairns, P. (2019). *Doing better statistics in human-computer interaction*. Cambridge University Press.
 - ▶ Ch15: What Makes a Good Likert Item?
 - ▶ Ch16: The Meaning of Factors
 - ▶ Ch17: Unreliable Reliability: The Problem of Cronbach's Alpha
 - ▶ Ch18: Tests for Questionnaires